

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a6bdbc1ede4503298.jsonl`
- SHA-256: `30c22b1a46b66944a219c084f4db317f5b21e6689880328c629a15bb8e0fccc7`
- Source modified: `2026-06-16T03:21:40+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:20:17.540Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://www.mdpi.com/1424-8220/24/13/4372>

****Title:**** Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera (Sensors 2024)

****Found via:**** Shuttle/ball trajectory ? rally/point segmentation in racket sports search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:20:19.622Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:20:25.463Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

4. assistant (2026-06-16T03:20:28.400Z)

The MDPI page returned 403. Let me try alternative approaches ? the PubMed Central mirror or a Google cache, and a web search for the paper's content.

5. user (2026-06-16T03:20:35.601Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/?term=Enhancing+Badminton+Game+Analysis+Shot+Refinement+Shuttlecock+Tracking+Hit+Detection>

Redirect URL: <https://pmc.ncbi.nlm.nih.gov/articles/?term=Enhancing+Badminton+Game+Analysis+Shot+Refinement+Shuttlecock+Tracking+Hit+Detection>

Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "https://pmc.ncbi.nlm.nih.gov/articles/?term=Enhancing+Badminton+Game+Analysis+Shot+Refinement+Shuttlecock+Tracking+Hit+Detection"

- prompt: "Find the article "Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera" and report any PMC ID or link."

6. user (2026-06-16T03:20:41.689Z)

Web search results for query: "Enhancing Badminton Game Analysis Shot Refinement Fusion Shuttlecock Tracking Hit Detection Monocular Camera Sensors 2024 method accuracy"

Links: [{"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Future Prediction of Shuttlecock Trajectory in Badminton Using Player's Information - PubMed", "url": "https://pubmed.ncbi.nlm.nih.gov/37233318/"}, {"title": "Fine reconstruction of badminton swing dynamic trajectory assisted by event camera | Scientific Reports", "url": "https://www.nature.com/articles/s41598-026-46443-8"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "(PDF) Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.researchgate.net/publication/382039345_Enhancing_Badminton_Game_Analysis_An_Approach_to_Shot_Refinement_via_a_Fusion_of_Shuttlecock_Tracking_and_Hit_Detection_from_Monocular_Camera"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"}, {"title": "A Model for Analyzing Shuttlecock Landing Position Using Real-Time Object Detection Techniques | Journal of Applied Science and Emerging Technology", "url": "https://ph01.tci-thaijo.org/index.php/JASCI/article/view/265289"}]

Based on the search results, I found comprehensive information about the research you're looking for:

Study Overview

The study proposes a novel method to extract shots in badminton games from a monocular camera. The research was published in 2024 by authors Yi-Hua Hsu, Chih-Chang Yu, and Hsu-Yung Cheng in the journal *Sensors*.

Methodology

The approach combines two main techniques:

1. TrackNet, a deep neural network designed for tracking small objects, is used to extract the flight trajectory of the shuttlecock.
2. The YOLOv7 model is used to identify whether the player is swinging.

A shot refinement algorithm was proposed to obtain the correct hitting moment, as both TrackNet

and YOLOv7 may have detection misses and false detections.

Method Accuracy

The proposed method achieved strong performance metrics: The method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches, with 1582 rallies of the Badminton World Federation (BWF) match videos.

This is a significant improvement compared to the use of TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score. Additionally, the accuracy of shot type classification at three different thresholds is 72.1%, 65.4%, and 54.1%.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-06-16T03:20:44.298Z)

Found the PMC mirror. Let me fetch the full text from PMC for verbatim quotes.

8. user (2026-06-16T03:20:56.479Z)

Badminton Shot Analysis Paper Summary

Publication Details:

- **Journal:** Sensors (Basel, Switzerland)
- **Date:** July 5, 2024
- **Authors:** Yi-Hua Hsu, Chih-Chang Yu, Hsu-Yung Cheng
- **License:** Creative Commons Attribution (CC BY) 4.0

(1) Converting Shuttlecock Trajectory into Shot/Hit Segments

"TrackNet proposed the Direction Identification Method, which checks for sudden changes in the direction of the shuttlecock."

The paper uses "Peak Identification Method" and "Direction Identification Method" from TrackNet to detect direction changes in trajectories.

(2) Hit Detection: Combining Trajectory & Player Swing

"The objective of the proposed SRA is to combine the results of both modules, thereby providing the precise identification of the hit moment."

The system uses TrackNet for trajectory (HD-T) and YOLOv7 for action detection (HD-A), merging results to identify when players actually contact the shuttlecock.

(3) Shot Refinement Algorithm (SRA)

The algorithm addresses five cases:

Case 1: "set HD-Tt as HMT(k), where k is the same as the k of HSi(k)."

Case 4: "the frame exhibiting the highest confidence score within HSi(k) is selected as HMT(k) to account for the absent HD-T."

(4) Distinguishing In-Play from Dead-Time/Serve Detection

The paper does not explicitly address serve detection. It identifies rallies (in-play exchanges) but does not separately discuss dead-time classification.

(5) Accuracy/Recall/F1 Numbers & TrackNet Comparison

****Proposed SRA (tuned) at 0.5 IoU threshold:****

- "precision increasing to 0.843... recall rate of 0.882... and the F1 score reaching 0.862"
- Tuned version: "precision bounced back to 0.897, reflecting an increase of 37.3%"

****TrackNet alone:****

"precision of 0.588, a recall rate of 0.936, and an F1 score of 0.723 when the IoU threshold was set to 0.5"

(6) Dataset Size

"32 matches and 602 rallies across 8975 images" (BWF dataset)

"354 matches and 6855 rallies. Overall, the datasets included 16 players" (combined dataset)

(7) Trajectory Features for Classification

Seven-dimensional features used in shot-type classification:

- "Both players' ankle positions (x?, y?, x?, y?) at the hit moment"
- "Flight duration of the shuttlecock, denoted as T"
- "The displacement of the shuttlecock (?x, ?y)"

9. assistant (2026-06-16T03:21:00.017Z)

I have detailed extraction. Let me verify the key dataset/accuracy numbers, since the WebSearch summary and WebFetch gave slightly different figures (89.7% accuracy / 1582 rallies / 69 matches vs. 354 matches / 6855 rallies). Let me confirm the headline numbers and the dead-time aspect with one targeted fetch.

10. user (2026-06-16T03:21:05.561Z)

Performance Summary

****Abstract statement:****

"Our proposed method achieves an accuracy of 89.7%, a recall rate of 91.3%, and an F1 rate of 90.5% in 69 matches, with 1582 rallies of the Badminton World Federation (BWF) match videos."

****Comparison to TrackNet:****

"This is a significant improvement compared to the use of TrackNet alone, which yields 58.8% accuracy, 93.6% recall, and 72.3% F1 score."

****Shot type classification:****

"the accuracy of shot type classification at three different thresholds is 72.1%, 65.4%, and 54.1%."

11. assistant (2026-06-16T03:21:09.930Z)

I have verified the key numbers. The headline figures (89.7% / 91.3% / 90.5% on 69 matches, 1582 rallies) are the abstract's final reported numbers, and the TrackNet-alone comparison (58.8% / 93.6% / 72.3%) is confirmed. Now I'll provide the structured output.

This is primary research published in Sensors (MDPI), CC BY 4.0, by Hsu, Yu, and Cheng (July 2024). It is directly relevant to Dimension 2 (shuttle-trajectory ? rally signal processing, hit/contact detection via fusion of trajectory + swing).

12. user (2026-06-16T03:21:23.947Z)

Structured output provided successfully

13. assistant (2026-06-16T03:21:40.164Z)

Extracted and verified the source. Summary of what I found:

****Source:**** "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera," *Sensors* 2024, 24(13):4372, by Hsu, Yu & Cheng. Primary research, CC BY 4.0 (license-clean). Verified via the PMC mirror (PMC11244353) after the MDPI page returned 403.

****Relevance:**** Strongest fit for Dimension 2 (shuttle-trajectory ? rally signal processing). Key takeaway for the over-merge problem: trajectory direction-reversals/peaks mark hit events, and fusing the trajectory tracker with a YOLOv7 swing-action cue via a case-based Shot Refinement Algorithm lifts hit-moment F1 from 0.723 (TrackNet alone) to 0.905 ? a license-clean recipe for combining a strong shuttle tracker with a weaker kinematic cue, which is the boundary signal needed to split merged windows.

****Caveat flagged:**** the paper segments **shots within rallies**, not **rally start/end**; it does not model serve or dead-time, so it doesn't directly solve the rally-bridging root cause ? useful for boundary cues, not for the dead-time split itself.

One number discrepancy worth noting: the abstract's headline (89.7%/91.3%/90.5% on 69 matches, 1582 rallies) is what I cited; the body text reports per-IoU results (e.g., 0.897 precision / 0.882 recall / 0.862 F1 at 0.5 IoU) and several different dataset splits (e.g., a 32-match/602-rally BWF subset and a combined 354-match set), so the exact figures vary by table ? I used the abstract's final reported numbers as authoritative.